

We thank all reviewers. Most concerns are pre-empted in the paper’s objection section (A1–A6); below we point there and add only the new evidence requested.

A. Core contribution and factual points (R-XHFa)

Two of these points are factual and now corrected; the rest we address on their merits. The submission id is shown in the header, and “calibration across two backbones” misreads §5.3: the “re-fit” on ResNet, ViT, ConvNeXt and Swin is *independent per-backbone* refitting to demonstrate universality, not a combination across two. On *core contribution*, the paper offers (i) the *QAC/QCDI evaluation protocol* for quality-conditioned calibration, our main object of study; (ii) the *quality taxonomy* (§3.2) sorting backbones into repairable tiers (Std VIB = Quality-Fragile, $0 < \text{QCDI} \leq 0.10$); and (iii) *Prop 1* (§4), an impossibility result showing a *scalar* TS cannot undo a rank reversal, so a continuous slope is the minimal sufficient repair. The 0.45/0.50 thresholds are not load-bearing: reversal holds across $\tau \in [0.40, 0.50]$ (§A10). Finally, ITB deliberately spans multiple degradations (blur, colour-temperature, brightness, contrast, completeness) so that QCDI probes calibration under heterogeneous quality shift, not a single artifact.

B. The mechanism is not specific to the weak Std VIB (R-isv7, R-8hpP)

Three points answer this. (i) Std VIB is *by design* the Quality-Fragile repairable prototype ($0 < \text{QCDI} \leq 0.10$) of our taxonomy (§3.2); we deliberately pick a repairable-tier representative to demonstrate the fix, whereas an already quality-aware backbone (ResNet-50) needs no repair (A4, Table 3). Its low AUC is the defining property of the sample, not a flaw. (ii) The mechanism is not confined to Std VIB: using an *external* corruption-severity scalar as $\bar{q}$ (not our IQA module), on both ResNet-50 and ViT-Tiny QCTS makes ρ more negative on 18/18 corruptions (a mechanism probe: the fix engages from an external signal without harming these already-calibrated backbones), while scalar TS is inert ($|\Delta\rho| < 10^{-4}$, 18/18), a monotone-invariant clean control. A Wilcoxon signed-rank test over these 18 corruptions confirms significance ($p = 3.8 \times 10^{-6}$). Where there is headroom the gain is sizeable (ViT-Tiny ρ from -0.139 to -0.210). (iii) On strong, quality-aware backbones (Table 3; ViT-Tiny/ConvNeXt/Swin) TS induces the QCDI sign-flip ($+0.023/ +0.136/ +0.020$ to $-0.029/ -0.022/ -0.021$) and QCTS pulls QCDI back *toward zero*. For the headline Std VIB case this is achieved by *improving* the low-quality stratum, not degrading the high-quality one: ECE-LQ falls $0.146 \rightarrow 0.079$ while ECE-HQ also improves $0.129 \rightarrow 0.075$ (Table 1).

C. Stability, baselines, and real data

T_0/α **stability (R-isv7)**. α sits in a flat NLL basin (§A18: $\alpha \in \{0.34, 0.96, 0.37\}$ within $\Delta\text{NLL} < 0.002$), yet the delivered calibration stays low: across 500 bootstrap refits per backbone (ResNet/ConvNeXt/ViT/Swin) the ECE-LQ never exceeds 0.058 (means 0.037–0.051; worst-case 95% upper bound 0.058 on Swin-Tiny), consistent with Prop 1, under which any nonzero slope suffices. The loose identifiability of α thus reflects a flat objective, not a fragile calibrator.

Re-marginalisation baseline (R-8hpP). It is present: the deployment-time re-marginalised baseline is §5.2 Eq.(5), ρ_a (MC-marginalised, $N_{\text{MC}}=20$) = -0.163 vs ρ_b (deterministic- μ) = $+0.241$, a 100% flip; the full table is §A20.

Real low-quality data (R-isv7, R-8hpP). QCTS is not evaluated only on synthetic degradation. As pre-empted in A1, we ran EfficientNet-B3 on 174 real-world low-quality dermoscopy images and recovered ECE 0.073 $[0.038, 0.125]$ vs. ITB-LQ’s 0.146 (supp real-LQ table). Directly applying QCTS to a real *paired* malignant cohort is future work, since no such cohort is publicly available, so the mechanism, not the synthetic pipeline, is the claim.

IQA module (R-isv7, R-XHFa). The 5-head module is validated in supp §A2 (mean SRCC 0.895); the mean-of-5 scalar suppresses noise, and expert-rated validation is a stated limitation.

D. Scope (R-8hpP, R-6xgi)

We thank R-8hpP(5) and R-6xgi(4); the Std VIB point is scope, not soundness (A4, Table 3). The headline *magnitude* (73% QCDI reduction, halved LQ ECE) is Std-VIB-specific, whereas the reversal-mechanism *direction* holds across all five backbones and 18 ImageNet-C corruptions (above); the contribution does not hinge on the weak backbone. In the camera-ready we will state this magnitude explicitly as a Quality-Fragile-prototype illustration and lead with the direction-level result across five backbones and 18 corruptions.